

Logistic Regression

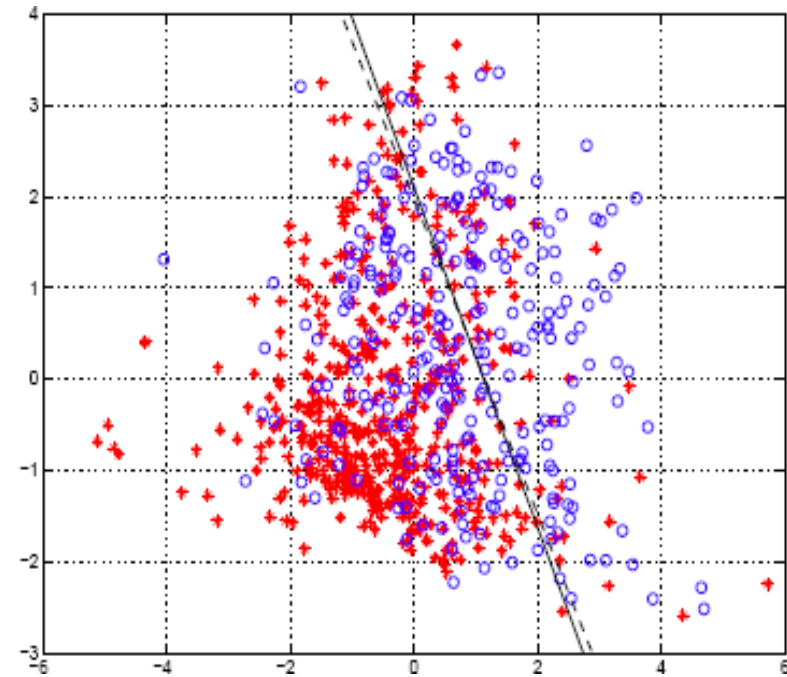
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Classification Based on Probability

- Instead of just predicting the class, give the probability of the instance being that class
 - i.e., learn $p(y | \mathbf{x})$
- Comparison to perceptron:
 - Perceptron doesn't produce probability estimate
 - Perceptron (and other discriminative classifiers) are only interested in producing a discriminative model
- Recall that:

$$0 \leq p(\text{event}) \leq 1$$

$$p(\text{event}) + p(\neg \text{event}) = 1$$



Logistic Regression

- Takes a probabilistic approach to learning discriminative functions (i.e., a classifier)

- $h_{\theta}(\mathbf{x})$ should give $p(y=1 \mid \mathbf{x}; \boldsymbol{\theta})$

Want $0 \leq h_{\theta}(\mathbf{x}) \leq 1$

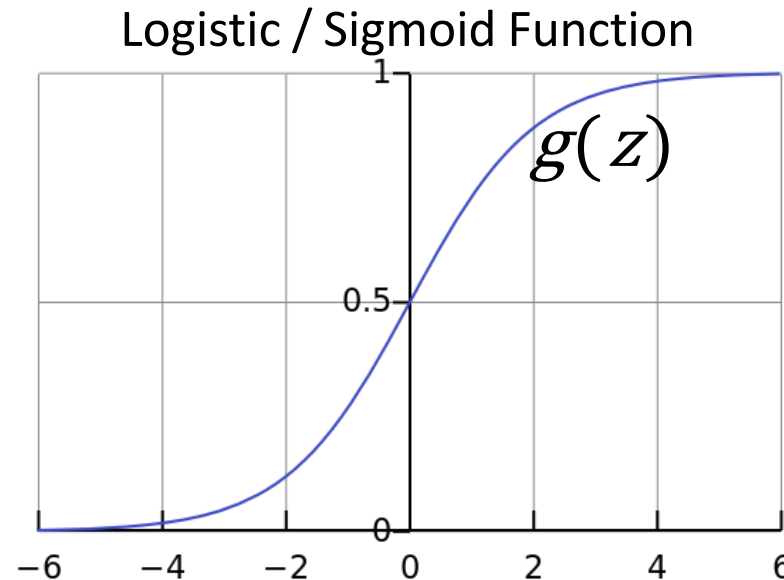
Can't just use linear regression with a threshold

- Logistic regression model:

$$h_{\theta}(\mathbf{x}) = g(\boldsymbol{\theta}^T \mathbf{x})$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

$$h_{\theta}(\mathbf{x}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}}}$$



Interpretation of Hypothesis Output

$h_{\theta}(\mathbf{x}) = \text{estimated } p(y=1 \mid \mathbf{x}; \theta)$

Example: Cancer diagnosis from tumor size

$$\mathbf{X} = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumor Size} \end{bmatrix}$$

$$h_{\theta}(\mathbf{x}) = 0.7$$

→ Tell patient that 70% chance of tumor being malignant

Note that: $p(y=0 \mid \mathbf{x}; \theta) + p(y=1 \mid \mathbf{x}; \theta) = 1$

Therefore, $p(y=0 \mid \mathbf{x}; \theta) = 1 - p(y=1 \mid \mathbf{x}; \theta)$

- Equivalently, logistic regression assumes that

$$\log \frac{p(y=1 \mid \mathbf{x}; \theta)}{p(y=0 \mid \mathbf{x}; \theta)} = \theta_0 + \theta_1 x_1 + \dots + \theta_d x_d$$

Side Note:

the odds in favour of an event is the quantity $p/(1-p)$, where p is the probability of the event

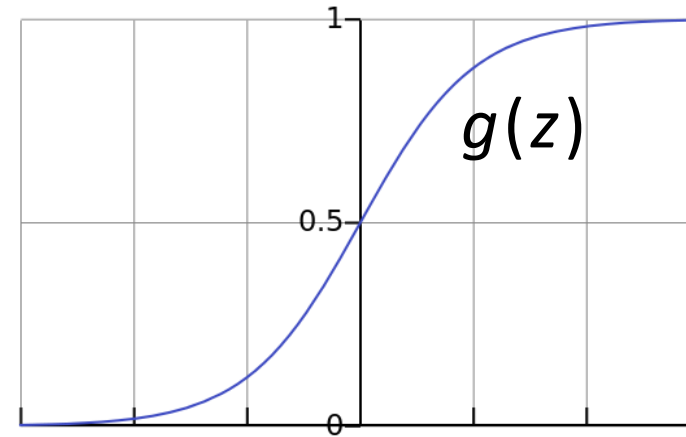
E.g., If I toss a fair dice, what are the odds that I will have a 6?

- In other words, logistic regression assumes that the log odds is a linear function of $\mathbf{x}$

Logistic Regression

$$h_{\theta}(\mathbf{x}) = g(\theta^T \mathbf{x})$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

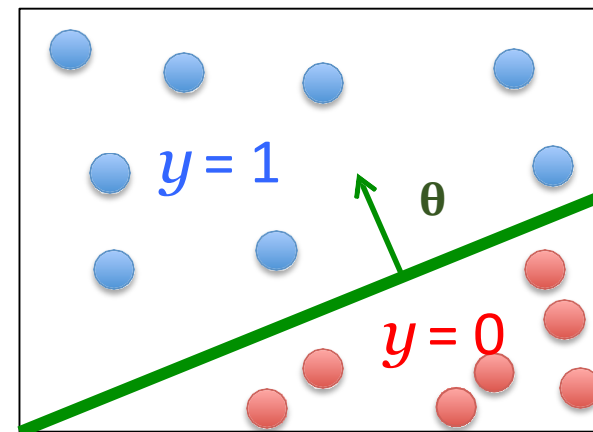


$\theta^T \mathbf{x}$ should be large negative values for negative instances

$\theta^T \mathbf{x}$ should be large positive values for positive instances

Assume a threshold and ...

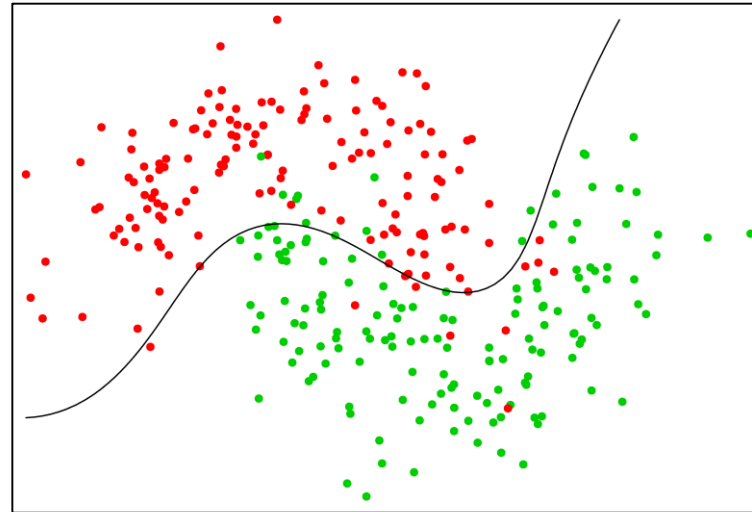
- Predict $y=1$ if $h_{\theta}(\mathbf{x}) \geq 0.5$
- Predict $y=0$ if $h_{\theta}(\mathbf{x}) < 0.5$



Non-Linear Decision Boundary

- Can apply basis function expansion to features , same as with linear regression

$$\mathbf{X} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_1 x_2 \\ x_1^2 \\ x_2^2 \\ x_1^2 x_2 \\ x_1 x_2^2 \\ \vdots \\ \vdots \end{bmatrix}$$



Logistic Regression

- Given $\{ (\mathbf{x}^{(1)}, y^{(1)}), (\mathbf{x}^{(2)}, y^{(2)}), \dots (\mathbf{x}^{(n)}, y^{(n)}) \}$

Where $\mathbf{x}^{(i)} \in \mathbb{R}^d, y^{(i)} \in \{0, 1\}$

- Model: $h_{\theta}(\mathbf{x}) = g(\boldsymbol{\theta}^T \mathbf{x})$

$$g(z) = \frac{1}{1+e^{-z}}$$

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}$$

$$\mathbf{x}^T = [1, x_1 \dots x_d]$$

Objective Function

Can't just use squared loss as in linear regression:

$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^n (h_{\theta}(\mathbf{x}^{(i)}) - y^{(i)})^2$$

Using the logistic regression model

$$h_{\theta}(\mathbf{x}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}}}$$

results in a non-convex optimization

Deriving the Cost Function via Maximum Likelihood Estimation

- Likelihood of data is given by: $l(\boldsymbol{\theta}) = \prod_{i=1}^n p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta})$
- So , looking for the $\boldsymbol{\theta}$ that maximizes the likelihood

$$\boldsymbol{\theta}_{\text{MLE}} = \arg \max_{\boldsymbol{\theta}} l(\boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta}} \prod_{i=1}^n p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta})$$

- Can take the log without changing the solution:

$$\boldsymbol{\theta}_{\text{MLE}} = \arg \max_{\boldsymbol{\theta}} \log \prod_{i=1}^n p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta})$$

$$= \arg \max_{\boldsymbol{\theta}} \sum_{i=1}^n \log p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta})$$

Expand as follows:

$$\boldsymbol{\theta}_{\text{MLE}} = \arg \max_{\boldsymbol{\theta}} \sum_{i=1}^n \log p(y^{(i)} \mid \mathbf{x}^{(i)}; \boldsymbol{\theta})$$

$$= \arg \max_{\boldsymbol{\theta}} \sum_{i=1}^n [y^{(i)} \log p(y^{(i)} = 1 \mid \mathbf{x}^{(i)}; \boldsymbol{\theta}) + (1 - y^{(i)}) \log(1 - p(y^{(i)} = 1 \mid \mathbf{x}^{(i)}; \boldsymbol{\theta}))]$$

Logistic regression objective: $\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$

$$J(\boldsymbol{\theta}) = - \sum_{i=1}^n [y^{(i)} \log h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) + (1 - y^{(i)}) \log(1 - h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}))]$$

Intuition Behind the Objective

$$J(\boldsymbol{\theta}) = - \sum_{i=1}^n [y^{(i)} \log h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) + (1 - y^{(i)}) \log(1 - h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}))]$$

Cost of a single instance:

$$\text{Cost}(h_{\boldsymbol{\theta}}(\mathbf{x}), y) = \begin{cases} -\log(h_{\boldsymbol{\theta}}(\mathbf{x})) & \text{if } y = 1 \\ -\log(1 - h_{\boldsymbol{\theta}}(\mathbf{x})) & \text{if } y = 0 \end{cases}$$

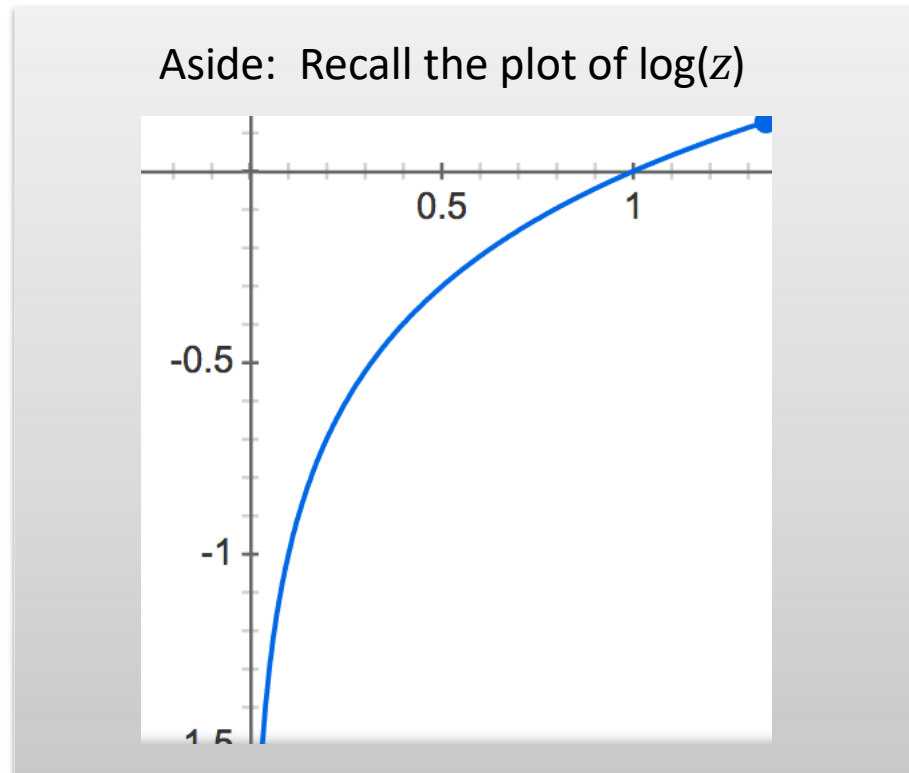
Can re-write objective function as

$$J(\boldsymbol{\theta}) = \sum_{i=1}^n \text{cost}(h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}), y^{(i)})$$

Compare to linear regression: $J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^n (h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)})^2$

Intuition Behind the Objective

$$\text{Cost}(h_{\theta}(\mathbf{x}), y) = \begin{cases} -\log(h_{\theta}(\mathbf{x})) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(\mathbf{x})) & \text{if } y = 0 \end{cases}$$

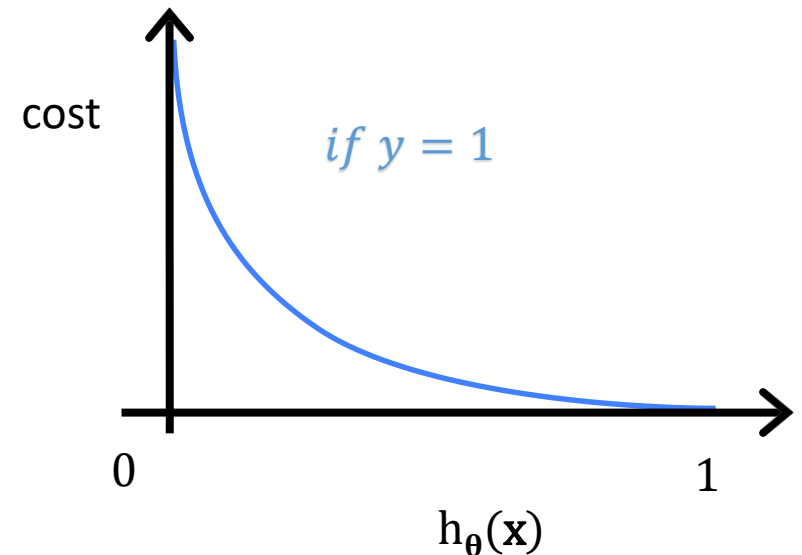


Intuition Behind the Objective

$$\text{Cost}(h_{\theta}(\mathbf{x}), y) = \begin{cases} -\log(h_{\theta}(\mathbf{x})) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(\mathbf{x})) & \text{if } y = 0 \end{cases}$$

If $y=1$

- Cost = 0 if prediction is correct
- As $h_{\theta}(\mathbf{x}) \rightarrow 0$, cost $\rightarrow \infty$
- Captures intuition that larger mistakes should get larger penalties
- e.g., predict $h_{\theta}(\mathbf{x}) = 0$, but $y = 1$

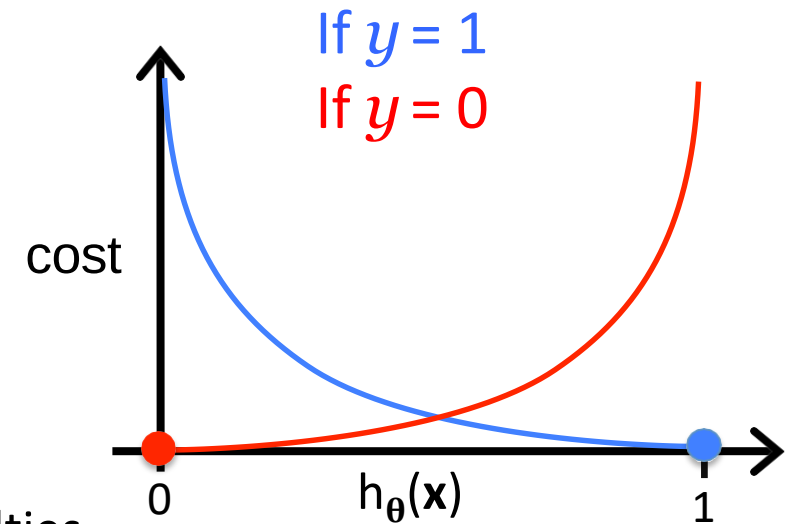


Intuition Behind the Objective

$$\text{Cost}(h_{\theta}(\mathbf{x}), y) = \begin{cases} -\log(h_{\theta}(\mathbf{x})) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(\mathbf{x})) & \text{if } y = 0 \end{cases}$$

If $y=0$

- Cost = 0 if prediction is correct
- As $(1 - h_{\theta}(\mathbf{x})) \rightarrow 0$, $\text{cost} \rightarrow \infty$
- Captures intuition that larger mistakes should get larger penalties



Regularized Logistic Regression

$$J(\theta) = - \sum_{i=1}^n [y^{(i)} \log h_{\theta}(\mathbf{x}^{(i)}) + (1 - y^{(i)}) \log(1 - h_{\theta}(\mathbf{x}^{(i)}))]$$

We can regularize logistic regression exactly as before:

$$\begin{aligned} J_{\text{regularized}}(\boldsymbol{\theta}) &= J(\boldsymbol{\theta}) + \frac{\lambda}{2} \sum_{j=1}^d \boldsymbol{\theta}_j^2 \\ &= J(\boldsymbol{\theta}) + \frac{\lambda}{2} \|\boldsymbol{\theta}_{[1:d]}\|_2^2 \end{aligned}$$

Gradient Descent for Logistic Regression

$$J_{\text{reg}}(\boldsymbol{\theta}) = - \sum_{i=1}^n [y^{(i)} \log h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) + (1 - y^{(i)}) \log(1 - h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}))] + \frac{\lambda}{2} \|\boldsymbol{\theta}_{[1:d]}\|_2^2$$

Want $\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$

- Initialize θ
- Repeat until convergence

$$\theta_j \leftarrow \theta_j - \alpha \frac{d}{d\theta_j} J(\boldsymbol{\theta}) \quad \begin{array}{l} \text{simultaneous update} \\ \text{for } j = 0, 1, \dots, d \end{array}$$

Use the naturel logarithm ($\ln = \log_e$) to cancel with the $\exp()$ in $h_{\boldsymbol{\theta}}(\mathbf{x})$

Gradient Descent for Logistic Regression

$$J_{\text{reg}}(\boldsymbol{\theta}) = - \sum_{i=1}^n [y^{(i)} \log h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) + (1 - y^{(i)}) \log(1 - h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}))] + \frac{\lambda}{2} \|\boldsymbol{\theta}_{[1:d]}\|_2^2$$

Want $\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$

Initialize $\boldsymbol{\theta}$

Repeat until convergence (Simultaneous update for $j = 0, 1, \dots, d$)

$$\theta_0 \leftarrow \theta_0 - \alpha \sum_{i=1}^n (h_{\boldsymbol{\theta}} \mathbf{x}^{(i)} - y^{(i)})$$

$$\theta_j \leftarrow \theta_j - \alpha \left[\sum_{i=1}^n (h_{\boldsymbol{\theta}} \mathbf{x}^{(i)} - y^{(i)}) x_j^{(i)} + \lambda \theta_j \right]$$

Simultaneous update for $j = 0, 1, \dots, d$

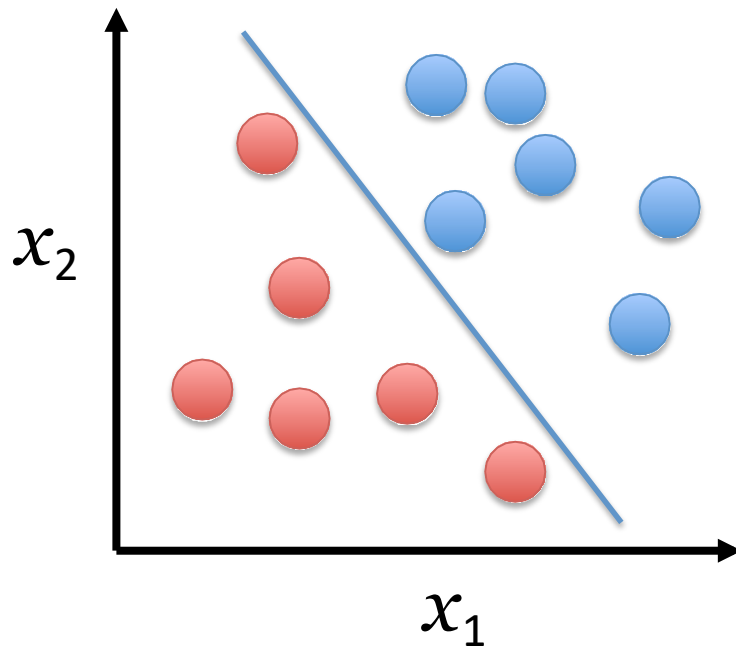
This looks identical to linear regression!!!

- Ignoring the $1/n$ constant
- However, the form of the model is very different:

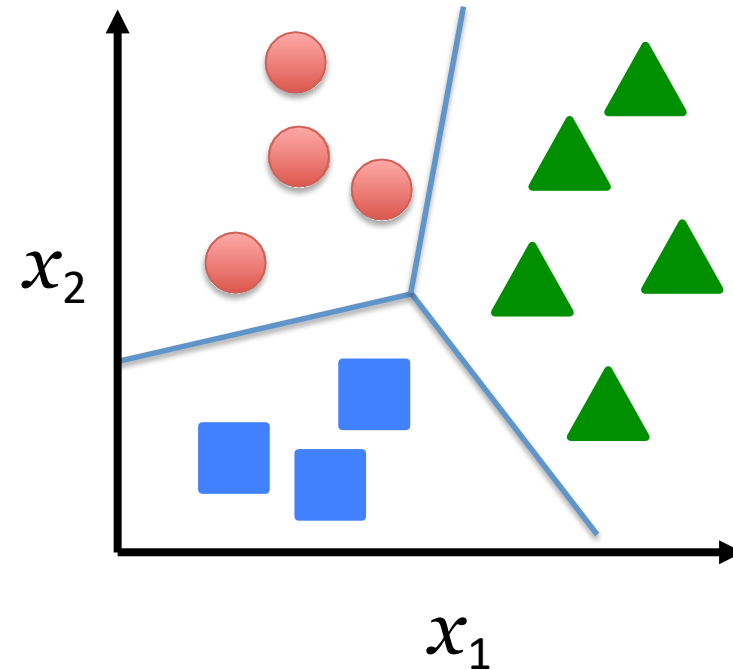
$$h_{\boldsymbol{\theta}}(\mathbf{x}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}}}$$

Multi-Class Classification

Binary classification:



Multi-class classification:



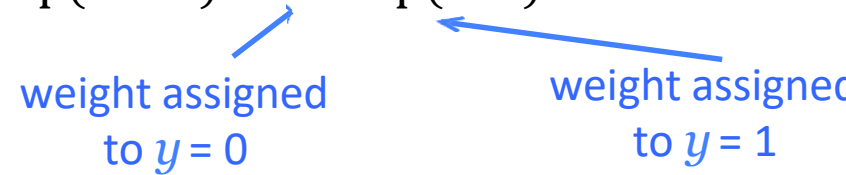
Disease diagnosis: healthy / cold / flu / pneumonia

Object classification: desk / chair / monitor / bookcase

Multi- Class Logistic Regression

- For 2 classes:

$$h_{\theta}(\mathbf{x}) = \frac{1}{1 + \exp(-\theta^T \mathbf{x})} = \frac{\exp(\theta^T \mathbf{x})}{1 + \exp(\theta^T \mathbf{x})}$$



weight assigned to $y = 0$ weight assigned to $y = 1$

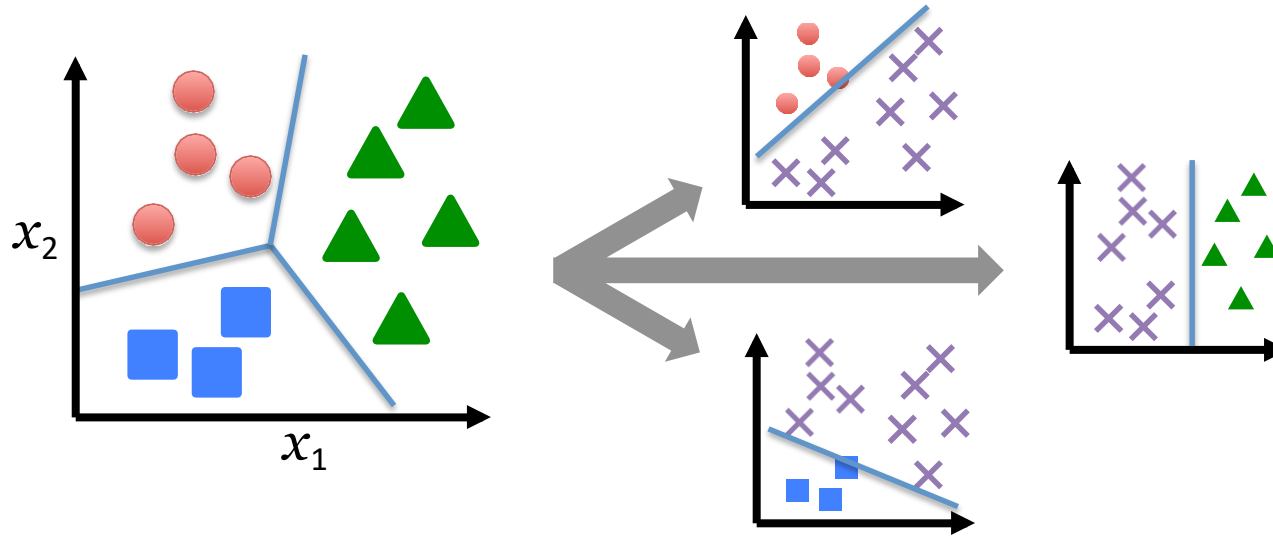
- For C classes $\{1, \dots, C\}$:

$$p(y=c \mid \mathbf{x}; \theta_1, \dots, \theta_c) = \frac{\exp(\theta_c^T \mathbf{x})}{\sum_{c=1}^C \exp(\theta_c^T \mathbf{x})}$$

- Called the **softmax** function

Multi-Class Logistic Regression

Split into One vs Rest:



- Train a logistic regression classifier for each class i to predict the probability that $y = i$ with

$$h_c(\mathbf{x}) = \frac{\exp(\boldsymbol{\theta}_c^T \mathbf{x})}{\sum_{c=1}^C \exp(\boldsymbol{\theta}_c^T \mathbf{x})}$$

Implementing Multi-Class Logistic Regression

- Use $h_c(\mathbf{x}) = \frac{\exp(\boldsymbol{\theta}_c^T \mathbf{x})}{\sum_{c=1}^C \exp(\boldsymbol{\theta}_c^T \mathbf{x})}$ as the model for class c
- Gradient descent simultaneously updates all parameters for all models
 - Same derivation as before, just with the above $h_c(\mathbf{x})$
- Predict class label as the most probable label

$$\max_c h_c(\mathbf{x})$$

Thank You